

The Four-Layer AI Visibility Framework

How Language Models Encode, Prefer, Retrieve, and Reason About Business Entities

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Abstract

As large language models (LLMs) become primary information intermediaries in commercial decision-making, the mechanisms by which these models acquire, retain, and surface knowledge about business entities have significant practical and theoretical implications. This paper introduces the **Four-Layer AI Visibility Framework**, a structured analytical model that decomposes AI-mediated entity visibility into four distinct operational layers: (1) pre-training corpus representation, (2) post-training reinforcement learning preference installation, (3) real-time retrieval architecture, and (4) large reasoning model chain-of-thought integration. We argue that existing practitioner frameworks for Generative Engine Optimization (GEO) and Answer Engine Optimization (AEO) address only the third layer, leaving three foundational mechanisms largely unaddressed. We further examine platform-specific retrieval architectures across Perplexity AI, Anthropic Claude, OpenAI ChatGPT, and Google Gemini, and discuss the under-researched implications of dialectal variation in Spanish-language AI responses for Latin American B2B entities. The framework is intended to serve as both a theoretical model for AI visibility research and a practitioner guide for organizations seeking to improve their representation in LLM-generated responses.

Keywords: large language models, AI visibility, generative engine optimization, answer engine optimization, entity representation, retrieval-augmented generation, reinforcement learning from human feedback, reasoning models, knowledge graphs, Wikidata, Spanish NLP, dialectal variation, B2B marketing, Latin America

1. Introduction

The widespread deployment of large language models as consumer and enterprise information tools has created a new class of information intermediary distinct from traditional search engines. When users query systems such as ChatGPT (OpenAI), Claude (Anthropic), Gemini (Google), or Perplexity AI, the responses they receive are not ranked lists of external documents but synthesized natural language outputs that directly name, describe, evaluate, and recommend specific entities — including businesses, products, services, and individuals.

This shift has material implications for how organizations are perceived and discovered in commercial contexts. Research from Conductor (2026), based on analysis of approximately 17 million AI-generated responses, identifies ChatGPT as responsible for approximately 70% of AI-originated referral traffic, with

Perplexity growing over 200% in 2025. The market for AI visibility optimization tooling was estimated at USD 2.37 billion in 2025, growing at 16% annually (Grand View Research, 2025).

Despite this commercial significance, the academic literature on AI-mediated entity visibility remains limited. Existing practitioner frameworks — broadly labeled GEO (Generative Engine Optimization) or AEO (Answer Engine Optimization) — focus predominantly on content-level interventions designed to influence real-time retrieval systems. These frameworks implicitly treat AI model responses as analogous to search engine results pages, suggesting that the primary mechanism of influence is content optimization for retrieval ranking.

We argue this framing is insufficient. LLM responses are determined by at least four mechanistically distinct processes, each operating at a different stage of the model lifecycle and each requiring distinct intervention strategies. Conflating these processes leads to misallocated optimization effort and incomplete visibility outcomes.

This paper makes the following contributions:

- We introduce the **Four-Layer AI Visibility Framework**, which systematically distinguishes the mechanisms by which LLMs acquire and surface entity knowledge.
- We provide a comparative analysis of real-time retrieval architectures across the four dominant AI response platforms as of February 2026.
- We identify large reasoning models (LRMs) as a newly relevant visibility layer with distinct optimization implications.
- We highlight dialectal variation in Spanish as an under-researched source of systematic visibility disparity for Latin American entities.

2. Background and Related Work

2.1 Search Engine Optimization and Its Limitations as an AI Visibility Model

Traditional Search Engine Optimization (SEO) involves interventions targeting the indexing, ranking, and presentation of web content in response to user queries. Core mechanisms include technical site structure, content quality signals, inbound link authority, and query-content relevance matching (Brin & Page, 1998; Cutts, 2012).

The analogy to AI response optimization is superficially compelling: both involve influencing how an information system surfaces an entity in response to a user query. However, the mechanisms differ fundamentally. Search engines retrieve pre-existing documents; language models generate novel text. Search engines operate in real time against live indices; language models carry fixed parametric knowledge from training that is only partially supplemented by retrieval. Search rankings are explicit and auditable; LLM response tendencies are emergent from training dynamics and substantially opaque.

2.2 Emerging GEO/AEO Literature

The practitioner literature on GEO and AEO has grown rapidly since 2024. Aggarwal et al. (2023) introduced the term "Generative Engine Optimization" and identified content-level factors — including statistics, quotations, and fluency — that increased citation rates in early retrieval-augmented generation (RAG) systems. Subsequent work has examined prompt-level interventions (Liu et al., 2024), structured

data markup effects (Chen & Park, 2025), and platform-specific citation patterns.

This body of work is valuable but narrow: it addresses what corresponds to Layer 3 in our framework (real-time retrieval) and largely ignores the pre-training, post-training, and reasoning layers. It is also predominantly English-language, creating a significant research gap for non-English markets.

2.3 Entity Representation in Language Model Training

A parallel literature in NLP addresses how named entities are represented in language model weights. Petroni et al. (2019) demonstrated that pre-trained language models encode factual associations as implicit "knowledge bases," with recall accuracy correlating with entity frequency in training data. Mallen et al. (2022) showed that entity popularity — measured by Wikipedia page views as a proxy for training corpus presence — predicts factual accuracy in LLM outputs. Roberts et al. (2020) examined how fine-tuning affects pre-trained factual knowledge.

This literature establishes the theoretical basis for Layer 1 of our framework but has not been applied to the commercial question of how organizations can improve their training corpus representation.

2.4 Reinforcement Learning and Model Preferences

The effects of RLHF (Reinforcement Learning from Human Feedback) on model outputs are well-documented in terms of safety and helpfulness improvements (Ouyang et al., 2022; Bai et al., 2022). Less studied are the effects of RLHF on entity citation preferences — specifically, whether the preference installation process systematically advantages certain types of sources, domains, or content styles. Our framework identifies this as a distinct and actionable visibility layer.

2.5 Large Reasoning Models

The deployment of chain-of-thought reasoning capabilities in models such as OpenAI's o1/o3 series, Anthropic's Claude Opus, and DeepSeek-R1 introduces a novel visibility consideration: the model's internal reasoning process, not just its final response, is a site of entity citation and evidence integration. This layer has received no systematic treatment in the GEO/AEO literature to date.

3. The Four-Layer AI Visibility Framework

We define **AI visibility** as the probability that a given entity is accurately named, correctly described, and appropriately recommended by a language model in response to a relevant query. This probability is a function of four interacting layers.

3.1 Layer 1 — Pre-Training Corpus Representation

Definition: The degree to which an entity is present, frequently mentioned, and consistently represented in the text corpora used during initial model pre-training.

Mechanism: During pre-training, LLMs process massive text corpora and build parametric representations of world knowledge. Entity representation in this phase is determined by: (a) frequency of entity mentions across corpus sources; (b) consistency of entity naming and description; (c) authority weight of source documents; and (d) co-occurrence patterns with already-established entities.

Privileged sources: Based on documented model training data compositions and crawl patterns, the following source types appear to carry disproportionate weight: Wikipedia and Wikidata (structured

knowledge bases with citation requirements); academic preprint servers (SSRN, Zenodo, arXiv); professional directories (LinkedIn, Crunchbase, GitHub); and established media outlets with high domain authority.

The knowledge graph dimension: Wikidata's structured entity-relationship data (using properties such as P31/instance-of, P571/inception, P159/headquarters, P452/industry, P856/official website) provides a machine-readable entity schema that supplements the statistical representations learned from unstructured text. Entities with complete Wikidata entries are more likely to be represented accurately and with appropriate relational context.

The synthetic data loop: A notable 2025 development is the increasing volume of AI-generated content entering training corpora for subsequent model generations. This creates a potential reinforcement dynamic: entities frequently mentioned in AI-generated text may receive compounding representation advantages in future model training cycles.

Table 1: Primary AI web crawlers and associated models

Crawler identifier	Organization	Associated model(s)
GPTBot	OpenAI	GPT-4o, o1, o3 series
ClaudeBot	Anthropic	Claude 3.x, 4.x series
PerplexityBot	Perplexity AI	Perplexity
Google-Extended	Google DeepMind	Gemini series
Bingbot	Microsoft	ChatGPT (via Bing integration)

Access restrictions on these crawlers via robots.txt directives represent a first-order Layer 1 intervention point.

3.2 Layer 2 — Post-Training Reinforcement Learning Preference Installation

Definition: The systematic preferences for source types, content styles, and entity characteristics that are installed during post-training alignment processes.

Mechanism: After initial pre-training, LLMs undergo alignment fine-tuning using reinforcement learning techniques. These processes include RLHF (Reinforcement Learning from Human Feedback; Ouyang et al., 2022), RLAIF (Reinforcement Learning from AI Feedback; Bai et al., 2022), DPO (Direct Preference Optimization; Rafailov et al., 2023), and GRPO (Group Relative Policy Optimization; Shao et al., 2024). In all variants, the model learns to produce outputs that resemble high-reward responses in a training dataset of preference pairs.

Implications for entity visibility: Human and AI annotators systematically prefer responses that cite verifiable sources, include quantitative data, reference recognized entities with clear identity, maintain clear argumentative structure, and avoid hallucination. These preferences are transmitted to the model as generalized biases. The result is a systematic advantage for entities with strong Layer 1 representation and for content with explicit evidential structure.

Dialectal variation as a Layer 2 phenomenon: Prior research (Marco, 2025) demonstrated measurable differences in citation rates for content written in different Spanish variants. We hypothesize that this

reflects a composition effect in annotator demographics during RLHF: if annotators are not equally representative of all dialectal variants, the resulting preference model will encode dialectal biases that persist in deployed model behavior. This hypothesis requires further empirical validation.

3.3 Layer 3 — Real-Time Retrieval Architecture

Definition: The platform-specific mechanisms by which a model augments its response to a given query with content retrieved from external sources at inference time.

Mechanism: Retrieval-Augmented Generation (RAG; Lewis et al., 2020) architectures supplement LLM responses with documents retrieved from an external index. Each major AI response platform implements a distinct RAG architecture with different retrieval strategies, ranking models, and content selection criteria.

3.3.1 Perplexity AI

Perplexity is distinguished by its always-on retrieval architecture: every query triggers web search regardless of whether the model's training knowledge could suffice. Analysis of Perplexity's observable system behavior and practitioner-documented ranking patterns reveals a multi-stage pipeline.

Stage 1 — Initial retrieval: Candidate documents are retrieved from Perplexity's proprietary index (supplemented by Bing API) using query matching and basic relevance signals.

Stage 2 — Reranking: A reranking model re-scores candidates using factors including content freshness, domain authority, entity co-occurrence patterns, and topic multipliers. A documented quality threshold causes the entire candidate set to be discarded if too few results clear the minimum quality bar.

Stage 3 — Passage extraction and citation: Perplexity extracts and cites passages of approximately 40–60 words. This granularity means that page-level relevance is insufficient: the target passage must contain a direct, self-contained answer to the query.

Key practical findings from analysis of Perplexity's observed behavior:

- New content enters a trial window where early engagement determines survival in the index.
- Content freshness decays significantly within 2–3 days without updates.
- Certain domains receive algorithmic boosts: GitHub, LinkedIn, Reddit, Amazon, Coursera.
- Topic multipliers favor AI, technology, science, and business content.

3.3.2 Anthropic Claude

Claude's retrieval is powered by Brave Search, an independent index separate from Google. Empirical analysis shows approximately 87% overlap between Claude-cited sources and Brave's organic search index (vs. approximately 27% for ChatGPT with Bing), reflecting Brave's higher weight on independent link graphs and structured semantic data. Claude integrates search within its reasoning loop, potentially conducting multiple searches across a single response generation.

3.3.3 OpenAI ChatGPT

ChatGPT operates in two primary modes with distinct retrieval behaviors. In standard mode, the model draws primarily on training knowledge, with optional browsing supplementation. In Deep Research mode (released 2025), an agentic mode conducts dozens of sequential web searches, reads hundreds of source documents, and synthesizes findings across extended generation cycles. This mode rewards

comprehensive, long-form content with original data. ChatGPT's retrieval is Bing-powered, making Bing indexing and authority the primary Layer 3 optimization target for ChatGPT specifically.

3.3.4 Google Gemini

Gemini accesses Google's live search index including Google Scholar, Google News, and YouTube. E-E-A-T (Experience, Expertise, Authoritativeness, Trustworthiness) signals — Google's content quality framework — function as first-order signals in Gemini's retrieval architecture. Gemini is notably dominant in utilities and finance verticals (Conductor, 2026). YouTube content with structured transcripts receives elevated visibility in Gemini specifically.

3.4 Layer 4 — Large Reasoning Model Chain-of-Thought Integration

Definition: The presence and utility of entity content during the extended internal reasoning processes of large reasoning models (LRMs).

Mechanism: Large reasoning models such as OpenAI's o1/o3 series, Anthropic's Claude Opus, and DeepSeek-R1 produce extended chain-of-thought reasoning sequences before generating final responses. These sequences involve intermediate retrieval, evidence evaluation, and iterative synthesis. Under process reward model (PRM) training, each reasoning step is evaluated independently for quality.

Content that integrates naturally into reasoning chains shares several characteristics:

- **Unique, verifiable facts:** LRMs actively seek information not resolvable from general knowledge. Proprietary data, original measurements, and documented methodologies are structurally advantaged.
- **Explicit causal structure:** The reasoning process requires explanations of mechanisms, not just assertions. Content with explicit causal chains (Observation → Mechanism → Evidence → Implication) integrates naturally into reasoning sequences.
- **Quantified comparative claims:** Comparative reasoning steps in LRMs benefit from quantitative benchmarks that enable direct comparison without inference.
- **Entity density and cross-reference coherence:** When an LRM encounters multiple coherent resources from the same entity during a reasoning session — a study, a glossary, a website, a paper citation — the entity becomes contextually dense. This coherence effect amplifies effective visibility beyond any single content piece.

4. Platform–Layer Interaction Matrix

The four layers do not operate independently. Table 2 summarizes the primary interaction effects between layers.

Table 2: Cross-layer interaction effects on AI visibility

	L1 Pre-training	L2 Post-training RL	L3 Retrieval	L4 Reasoning
L1	—	Strong positive: high L1 presence reduces hallucination risk, increasing citation safety	Moderate positive: training familiarity may influence retrieval reranking	Strong positive: dense training knowledge supports reasoning confidence
L2	—	—	Moderate: annotator preferences may influence which retrieved sources get cited	Significant: GRPO training rewards correlate with evidence chain quality
L3	Weak (real-time retrieval feeds synthetic data loops)	None direct	—	Strong: LRM retrieval happens during reasoning, not just before
L4	—	—	Strong: LRMs conduct active retrieval within reasoning chains	—

5. The Dialectal Variation Problem in Spanish-Language Markets

The framework described above assumes, by default, that optimization variables are language-invariant. Prior research demonstrates this assumption fails for Spanish-language markets.

In a study analyzing 156 query-response pairs across Claude, ChatGPT, and Gemini (Marco, 2025), statistically significant differences in citation rates were observed for content written in Rioplatense Spanish, Mexican Spanish, Colombian Spanish, and neutral Spanish (as defined by the Fundación del Español Urgente normative standard). Citation rates for Rioplatense Spanish content were notably lower than for neutral Spanish content on queries oriented toward international or Latin American B2B contexts.

We hypothesize two contributing mechanisms:

- **Training corpus underrepresentation:** Rioplatense and regional Spanish variants may be systematically underrepresented relative to their speaker populations in the corpora used for pre-training major models, most of which were assembled with English-language data as the primary target.
- **Annotator demographics in RLHF:** If human annotators used in post-training are not demographically representative of Latin American Spanish speakers, the preference model may encode biases toward neutral or Peninsular Spanish as more authoritative.

Both mechanisms would produce the Layer 1 and Layer 2 effects we observe. Disentangling them requires further empirical work, which we identify as a priority research direction.

The practical implication is significant: B2B entities in Argentina, Uruguay, and other Rioplatense-speaking regions may face a systematic visibility disadvantage that cannot be addressed through Layer 3 optimization alone. Layer 1 interventions — specifically increasing training corpus representation alongside neutral Spanish content — and content strategy adjustments for Layer 2 alignment patterns are necessary.

6. Discussion

6.1 Implications for Entity Visibility Strategy

The Four-Layer Framework suggests a sequential strategic approach. Layer 1 interventions (Wikidata entity establishment, academic publication, crawl access) should precede Layer 3 optimizations, as strong Layer 1 representation creates compounding advantages across Layers 2, 3, and 4. Layer 3 optimizations are platform-specific and require distinct strategies for each of the four major platforms.

Layer 4 is the frontier. No established practitioner playbook exists for optimizing content for LRM reasoning integration. We propose that content structured as complete evidence chains — with explicit methodology, quantified claims, and causal structure — is most likely to function as effective reasoning-step evidence.

6.2 Limitations

This framework is based on analysis of publicly available documentation, observable model behavior, and prior research. The internal architectures of commercial AI systems are proprietary and may differ materially from documented or inferred characteristics. Layer 2 mechanisms (RLHF preference installation) are particularly opaque and difficult to study from the outside. Layer 4 mechanisms (LRM chain-of-thought behavior) are newly deployed and understudied. The dialectal variation findings are based on a sample of 156 query-response pairs and require replication at scale.

6.3 Future Research Directions

We identify the following as priority research directions:

- **Empirical measurement of training corpus representation** for B2B entities in Spanish-speaking markets, using entity popularity proxies (Wikipedia page views, Wikidata completion scores) as predictors of LLM citation accuracy.
- **Annotator demographics and RLHF dialectal bias:** A systematic study of how annotator language background affects RLHF preference models for Spanish-language content.
- **LRM chain-of-thought citation patterns:** Empirical analysis of which entity types are cited in extended thinking sequences for Spanish B2B queries.
- **Search index representation for LATAM entities:** A comparative analysis of how Latin American entities are represented in the Brave Search index (used by Claude), Google (used by Gemini), and Bing (used by ChatGPT), given documented differences in index composition.
- **Entity disambiguation in LATAM:** The prevalence of similar or identical company names in Latin American markets creates disambiguation challenges that may systematically affect entity visibility.

7. Conclusion

The mechanisms by which language models know, describe, and recommend business entities are more complex and more stratified than the dominant GEO/AEO practitioner frameworks acknowledge. This paper introduces the Four-Layer AI Visibility Framework as a more complete model, distinguishing pre-training corpus effects, post-training reinforcement learning preference installation, real-time retrieval architecture, and large reasoning model chain-of-thought integration.

Each layer requires distinct interventions and operates on a different timescale. Layer 3 interventions (content and technical optimization for retrieval) can show results within days. Layer 1 interventions (training corpus presence) operate on the timescale of model training cycles — months to years. Layer 2 effects are largely downstream of Layer 1 but may be partially addressed through content style alignment. Layer 4 is an emerging frontier requiring further research.

For organizations in Spanish-speaking markets, an additional dimension of complexity arises from dialectal variation effects that appear to operate through both Layer 1 and Layer 2 mechanisms. Addressing AI visibility in Latin American B2B contexts requires acknowledging and compensating for these systematic biases.

We offer this framework as a contribution to an emerging research area and invite replication, critique, and extension.

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Data Availability

The structured entity data referenced in this paper has been deposited in Wikidata as open-access records. The dialectal variation dataset underlying Section 5 is available from the corresponding author upon reasonable request. Wikidata identifiers for Exista.io's research portfolio are available at exista.io/research.

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